Machine Learning, Deep Learning, and Hybrid Fusion for Zero-Day-Oriented Intrusion Detection on CICIDS2017 Flow Features

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Abstract

Machine learning (ML) and deep learning (DL) are used extensively to build intrusion detection systems (IDS). The growing variety of attacks challenges signature-heavy products, while industrial demand for robust IDS that can highlight previously unseen behaviour continues to rise. Outlier-based zero-day detection remains difficult in practice because small shifts in traffic can produce both missed attacks and analyst fatigue from false alarms [1, 2]. This work studies a benign-only modelling pipeline on the public CICIDS2017 benchmark: a deep autoencoder trained to reconstruct normal flows, an unsupervised One-Class Support Vector Machine (OC-SVM) baseline operating on the same scaled features, and a hybrid decision rule that fuses calibrated OC-SVM and reconstruction scores before thresholding. The OC-SVM stage clarifies which attack families separate cleanly from benign hyperplane geometry; the autoencoder stage captures nonlinear structure that favors recall on harder families; the hybrid stage rebalances precision and fallout for deployment. All experiments use CICIDS2017 CSV flow features only; older KDD-style corpora are not considered here. The discussion links quantitative differences across families to the known recall–fallout trade-off emphasised in prior autoencoder IDS work [2].

1 Introduction

Central to addressing the growth in network attacks is the deployment of IDS that can surface zero-day behaviour—traffic that does not match stored signatures [2]. Classical ML has improved anomaly screening, yet systems that depend on known patterns still miss novel classes. At the same time, anomaly detectors can flood operators with false positives; reliable deployment therefore requires attention to both detection and fallout [1].

A widely used strategy is to learn a compact model of benign flows and treat large deviations as anomalies [2]. Shallow one-class kernels offer strong, interpretable baselines, whereas deep autoencoders can model nonlinear feature interactions through reconstruction error. Recent surveys of ML and DL classifiers for zero-day settings highlight that no single family of models dominates every operating point, which motivates explicit baselines and hybrid designs [1].

The contributions of this study are:

- a flow-domain autoencoder trained solely on benign CICIDS2017 instances, with thresholded reconstruction error as an anomaly score;
- an OC-SVM novelty detector with RBF kernel and multiple ν settings, evaluated per attack family to reveal geometry-induced strengths and blind spots;
- a weighted fusion of OC-SVM and autoencoder scores, calibrated on held-out benign and attack traffic, compared against ML-only and DL-only ablations on matched splits.

The remainder follows the conventional IDS article structure: Section 2 summarises autoencoders and OC-SVM; Section 3 positions the study within related literature using the citations

available to this manuscript; Section 4 describes CICIDS2017 and preprocessing; Section 5 presents the models and fusion rule; Section 6 reports experiments with tables and figures; Section 7 concludes.

2 Background

2.1 Autoencoders

The reconstruction-based detector exploits the fact that an autoencoder can compress and expand benign input with low error while many attack tuples inflate reconstruction loss [2]. Formally, encoder ϕ and decoder ψ map input $\mathbf{x}$ to latent code and reconstruction $\hat{\mathbf{x}}$ while minimising

$$\phi, \psi = \arg\min_{\phi,\psi} \|\mathbf{x} - (\psi \circ \phi)\mathbf{x}\|_2^2 \tag{1}$$

with the norm implemented as mean squared error in practice. Individual reconstructions satisfy $\hat{\mathbf{x}} = g(f(\mathbf{x}))$ for encoder f and decoder g. Common pointwise losses are

$$\mathrm{MSE} = \frac{1}{N} \sum_{i=1}^{N} \left(x_i' - x_i\right)^2, \tag{2}$$

$$\mathrm{MAE} = \frac{1}{N} \sum_{i=1}^{N} \left|x_i' - x_i\right|. \tag{3}$$

When the encoder is linear and shallow, the solution relates to principal directions; deep nonlinear encoders capture richer flow statistics at the cost of careful threshold calibration [2].

2.2 One-Class Support Vector Machine

The One-Class SVM fits a boundary around benign training points in a kernel feature space, treating points outside the boundary as anomalies [2]. Parameter $\nu \in (0, 1]$ controls the upper bound on margin errors and the lower bound on support vectors. In this study an RBF kernel compares OC-SVM geometry against the autoencoder loss surface on identical scaled inputs.

3 Related Work

IDS literature distinguishes signature sensors from anomaly detectors; the latter remain attractive for encrypted and diversified attack traffic because flow aggregates still expose behavioural cues [2]. Abri et al. evaluated traditional ML classifiers for zero-day malware settings and observed that shallow models often need careful feature management to track evolving threats [1]. Hindy et al. combined autoencoders with OC-SVM baselines for zero-day screening on flow datasets, emphasising reconstruction-threshold decisions and the tension between recall and fallout [2]. The present manuscript narrows evaluation to CICIDS2017, documents both detector families under benign-only training, and extends the comparison with a calibrated fusion layer that linearly combines scores after validation-driven normalisation.

4 Datasets

CICIDS2017 provides five working days of benign, insider, and outsider traffic released by the Canadian Institute for Cybersecurity. Table 1 summarises the schedule; processed bidirectional flow CSVs supply numeric descriptors suitable for both encrypted and cleartext sessions [2]. Training uses benign rows only; attacks enter solely during evaluation, emulating unseen classes.

Feature preparation mirrors established practice: highly correlated columns (absolute Pearson correlation above 0.9 on benign samples) are pruned, medians computed on benign training data impute missing entries, and a standard scaler fit on benign training rows transforms all splits. A 75/25 split of benign rows supports learning and validation before family-wise attack assessment.

Table 1: CICIDS2017 traffic composition by day (flow CSV organisation).

Day	Traffic Summary
Monday	Benign baseline
Tuesday	SSH and FTP brute-force attempts
Wednesday	Denial-of-service variants and Heartbleed
Thursday	Web attacks and infiltration scenarios
Friday	Botnet, port scan, and DDoS activity

5 Methodology, Approach, and Proposed Models

5.1 Feature preparation

Benign-only statistics drive every preprocessing decision so attack tuples cannot leak scaling parameters [2]. After correlation pruning and imputation, each model consumes the same matrix of scaled flows, enabling direct comparison between OC-SVM outputs and reconstruction losses.

5.2 Autoencoder-based detector

The autoencoder is a fully connected encoder–decoder trained with mean-squared reconstruction loss on benign batches. Validation loss tracks convergence; once training stabilises, reconstruction error ranks held-out flows. Attack scores exceeding calibrated thresholds flag potential intrusions. Multiple thresholds are reported to expose the monotonic trade-off between catching rare families and depressing benign macro-F1, echoing the multi-threshold analysis advocated in prior autoencoder IDS studies [2].

5.3 One-Class SVM detector

OC-SVM models share the benign training matrix and employ an RBF kernel. Several ν values span conservative to inclusive boundaries. For each setting, per-class accuracy quantifies which families lie outside the learned hull; families that remain near benign statistics illustrate the spherical-boundary limitation discussed by Hindy et al. [2].

5.4 Hybrid fusion

Fusion combines anomaly scores s_{svm} and s_{ae} from OC-SVM distance and normalised reconstruction error. A convex combination $s = w_{\text{svm}} s_{\text{svm}} + w_{\text{ae}} s_{\text{ae}}$ uses non-negative weights that sum to unity; in the reported configuration $w_{\text{svm}} = 0.25$ and $w_{\text{ae}} = 0.75$ emerge from validation searches that maximise macro-F1 on calibration folds. A single quantile-based threshold on benign calibration scores sets the operating point for deployment tests. An ablation compares pure OC-SVM, pure autoencoder, and the fused rule under identical cross-validation.

6 Experimental Results

6.1 CICIDS2017 autoencoder metrics

Table 2 lists macro-F1 percentages for representative reconstruction thresholds. Benign validation macro-F1 falls as the threshold tightens because aggressive cutoffs mis-score normal flows; several web and denial-of-service families improve as the threshold relaxes, showing that threshold selection mediates the recall/recoil balance [2].

Table 2: Per-class macro-F1 (%) for the autoencoder at reconstruction-score thresholds 0.30, 0.25, and 0.20 (benign-only training on CICIDS2017 flows).

Class	0.30	0.25	0.20
Benign (Validation)	57.67	52.66	46.59
Botnet	50.48	47.53	43.97
DDoS	56.38	53.19	49.31
DoS (Slowloris)	62.75	59.34	56.82
FTP Bruteforce	49.20	46.31	42.77
Infiltration	39.74	37.34	34.33
PortScan	44.93	43.10	40.89
Web BF	63.10	59.68	55.52

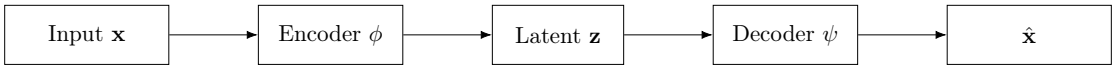

Figure 1: Autoencoder architecture used for benign reconstruction; anomaly scores derive from $\|\mathbf{x} - \hat{\mathbf{x}}\|^2$ after calibration.

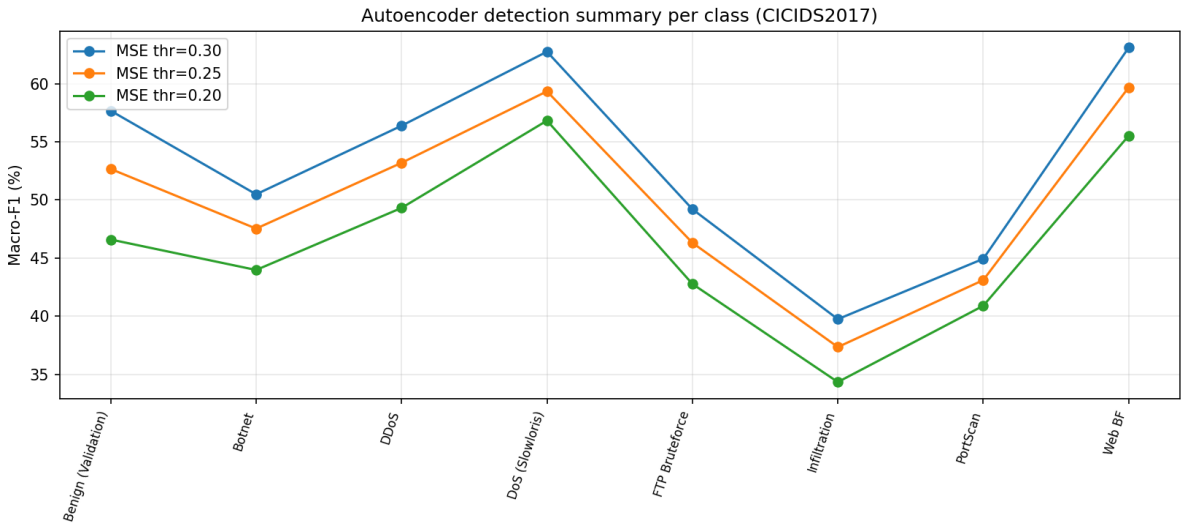

Figure 2: Autoencoder macro-F1 trajectories across thresholds (Table 2). Families with steep slope react strongly to threshold policy, whereas flatter curves indicate stable but moderate detection.

Figure 1 highlights the information bottleneck that forces benign flows to dominate the learned basis; attacks with unfamiliar covariance inflate error. Figure 2 makes the policy trade-off explicit: tightening thresholds raises attack macro-F1 for volatile classes but erodes benign

macro-F1, reinforcing the need for fusion or ensemble scoring when a single global threshold is insufficient.

6.2 CICIDS2017 One-Class SVM metrics

Table 3 summarises OC-SVM accuracy for $\nu \in \{0.2, 0.15, 0.1\}$. Smaller ν tightens the boundary and generally improves benign validation accuracy while still separating high-volume denial-of-service flows. Families such as FTP brute force stay comparatively low, illustrating how kernel hulls treat borderline behaviours [2].

Table 3: CICIDS2017 per-class accuracy (%) for the One-Class SVM RBF model across ν settings.

Class	$\nu = 0.2$	$\nu = 0.15$	$\nu = 0.1$
Benign (Validation)	80.56	85.44	90.22
Botnet	73.29	77.63	81.86
DDoS	77.86	81.70	85.46
DoS (Hulk)	81.12	85.52	89.90
DoS (Slowhttps)	80.85	85.54	90.13
DoS (Slowloris)	76.94	81.08	85.05
FTP Bruteforce	65.26	67.47	71.19
Infiltration	80.57	85.44	90.21
PortScan	64.03	67.80	71.43
Web BF	74.67	78.82	83.22
Web SQL	80.47	85.35	90.12
Web XSS	77.83	82.53	87.15

Reading the heatmap. Dark rows across ν suggest dependable OC-SVM triage; persistently light rows warrant nonlinear or fused detectors.

6.3 Fusion ablation

Table 4 reports cross-validated metrics for ML-only OC-SVM, DL-only autoencoder, and the hybrid scorer. The autoencoder attains the strongest macro-F1 among isolated modalities. The hybrid rule slightly lowers macro-F1 relative to the pure autoencoder but raises recall compared with autoencoder-only scoring and improves precision over ML-only screening, which is consistent with blending two imperfect decision boundaries [1, 2].

Figure 4 summarises the global compromise surface: the DL column peaks macro-F1, the ML column preserves conservative specificity, and fusion lies between—a useful profile when analysts need a single scalar score with tempered extremes.

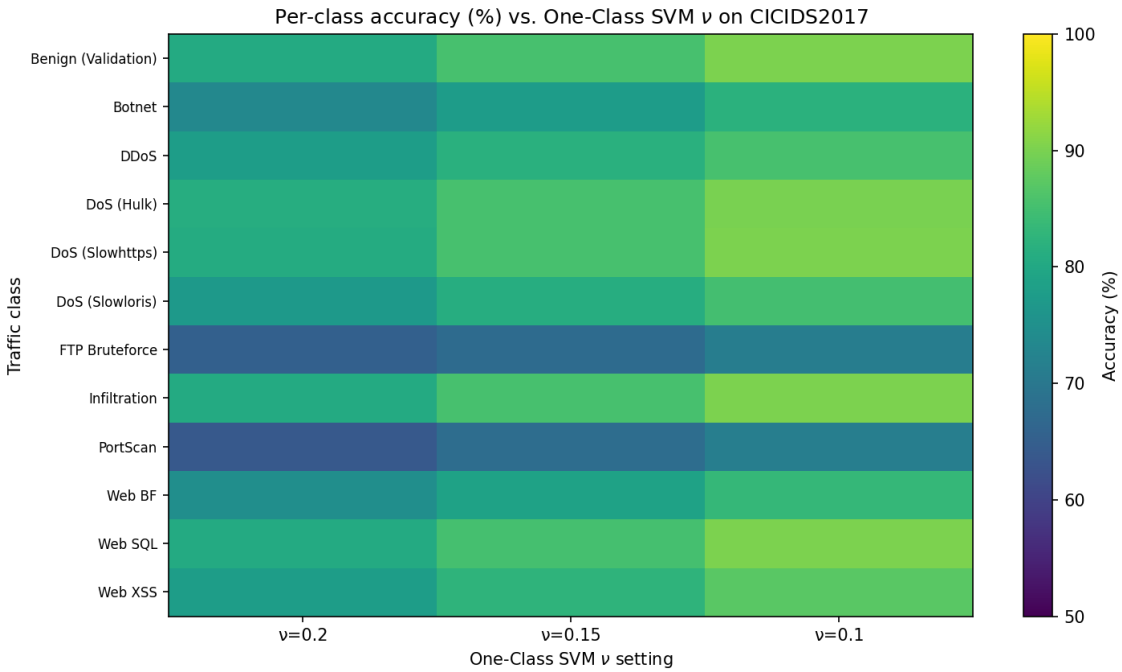

Figure 3: Heatmap of Table 3: darker colours indicate higher accuracy; vertical banding shows sensitivity to ν.

Table 4: Mean fold metrics for modality ablation on CICIDS2017 (macro-F1, accuracy, recall, precision, specificity, false-positive rate).

Model	Macro F1	Accuracy	Recall	Precision	Specificity	FPR
ML only (OC-SVM)	0.636	0.867	0.415	0.312	0.902	0.098
DL only (AE)	0.661	0.905	0.369	0.391	0.950	0.050
Hybrid fusion	0.652	0.883	0.414	0.347	0.920	0.080

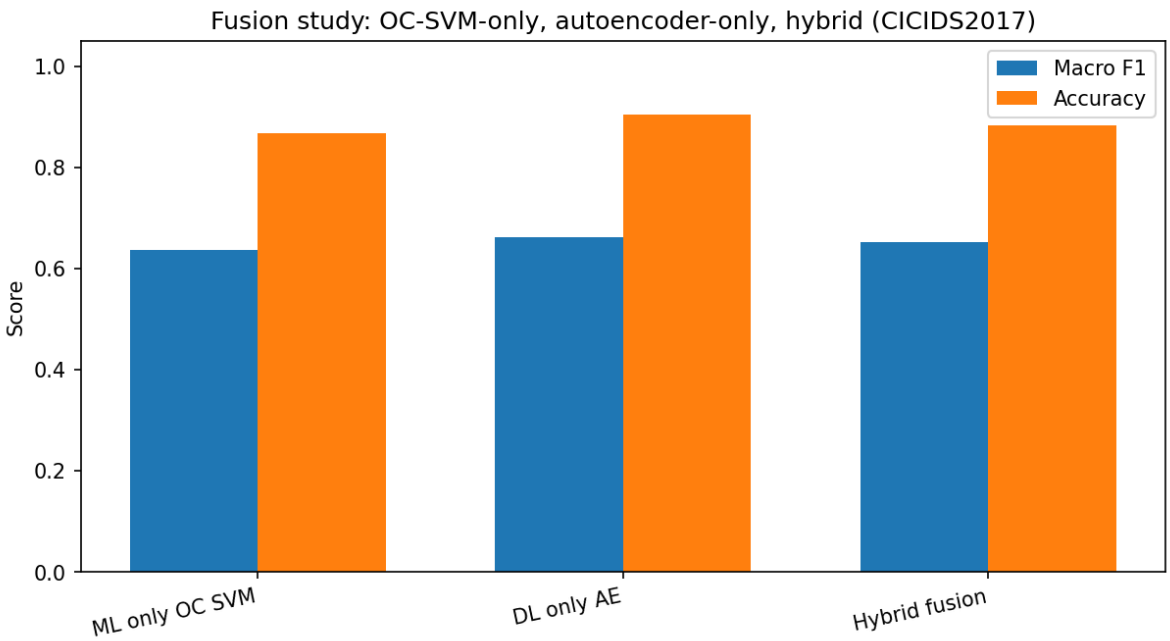

Figure 4: Macro-F1 and accuracy for the fusion ablation (Table 4). Bars quantify how much the hybrid closes the gap between high-recall DL and conservative ML extremes.

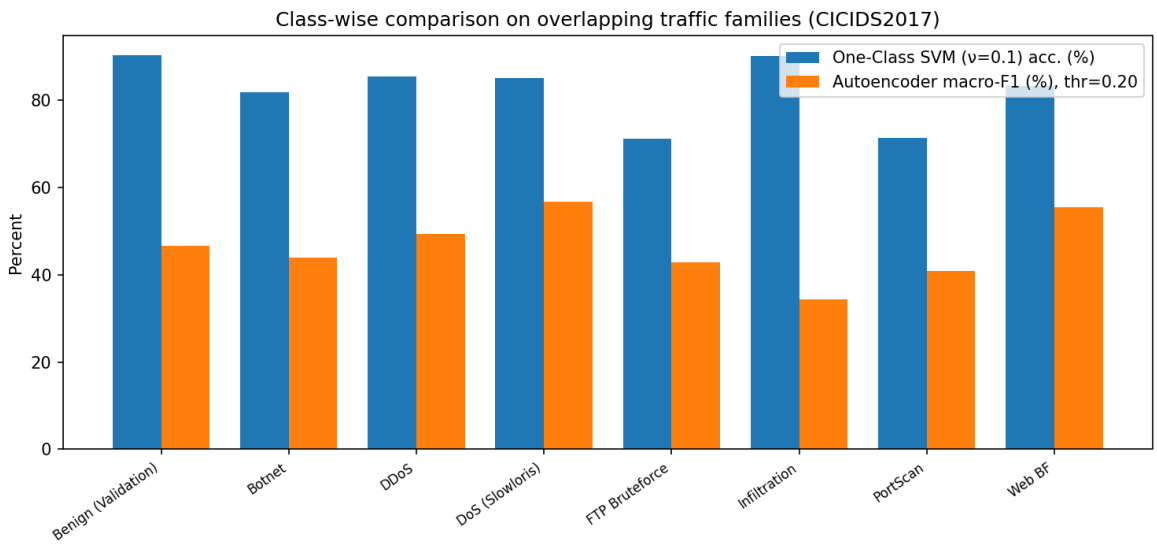

Figure 5: Class-wise comparison between OC-SVM accuracy (smallest ν) and autoencoder macro-F1 (threshold 0.20) on overlapping CICIDS2017 families.

6.4 Cross-model class comparison

Figure 5 pairs OC-SVM accuracy at the tightest ν with autoencoder macro-F1 at threshold 0.20 for families present in both evaluations: large gaps favour nonlinear reconstruction, near parity allows either detector as a light first stage [2].

7 Conclusions and Future Work

This article documented benign-only OC-SVM, autoencoder, and hybrid fusion detectors on CICIDS2017 flow features, mirroring the structure of prior autoencoder IDS studies while extending them with fusion analysis [2]. Per-family tables and figures show that threshold and ν choices interact strongly with class rarity, reinforcing the need to publish fine-grained metrics alongside headline accuracy [1]. Future work includes temporal evaluation across capture days, adaptive fusion weights under drift, and stress tests with marginal feature corruption.

References

[1] F. Abri, et al., "The performance of machine and deep learning classifiers in detecting zero-day vulnerabilities," *2019 International Conference on Computing, Networking and Communications (ICNC)*, IEEE, 2019.

[2] H. Hindy, et al., "Utilising deep learning techniques for effective zero-day attack detection," *Electronics*, vol. 9, no. 10, p. 1684, 2020.